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Statistical Arbitrage of Equity Index Volatility

A Machine Learning Approach to Dispersion Trading

MSc Risk Management & Financial Engineering | Applied Project | Quantitative Analysis

Nathan SEBBAG | Imperial College Business School | May 2026

Project Overview

What is Dispersion Trading?

The variance of an index is NOT the simple sum of component variances, it depends on their correlations.

$$\sigma_{Index}^2 = \sum_i \omega_i^2 \sigma_i^2 + \sum_i \sum_{j \neq i} \omega_i \omega_j \rho_{ij} \sigma_i \sigma_j$$

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Research Question

Can Random Matrix Theory-filtered correlation matrices and Deep Learning regime prediction improve the risk-adjusted performance of a systematic Dispersion strategy on the S&P 500?

1 The Trade

Long vol on SPX components · Short vol on SPX index · Delta & Vega neutral portfolio

2 The Edge

Market systematically overprices implied correlation · Exploitable Correlation Risk Premium (Driessen et al., 2009)

3 The Innovation

Random Matrix Theory to clean 50×50 correlation matrices · LSTM/XGBoost to predict regime shifts

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Methodology & Literature

01

Baseline Strategy

SPX Index vs. 50 Components Straddles
Delta-neutral rebalancing (Bloomberg API)

02

RMT Filtering

Noise isolation via Marchenko-Pastur law
Eigenvalue clipping on Σ_{50x50}

03

ML Signal Layer

LSTM: Predict implied vs. realized correlation spread
K-Means: Market regime clustering

04

Backtest Engine

Custom Python environment
Includes transaction costs & P&L attribution

Key Literature

FOUNDATIONS

— Driessen, Maenhout & Vilkov (2009) — Price of Correlation Risk

STOCHASTIC MODELS

— Bergomi (2015) — Stochastic Volatility Modeling

RMT & MATRICES

— Bouchaud & Potters — Financial Applications of RMT

ML IN FINANCE

— Gu, Kelly & Xiu (2020) — Asset Pricing via ML

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Timeline & Expected Contribution

MAY – JUNE	JUNE – JULY	JULY	JULY – AUG
Phase 1	Phase 2	Phase 3	Phase 4
Foundations	**RMT Layer**	**ML Signal**	**Write-up**
Bloomberg data extraction · Maths derivation (Itô P&L) · Baseline backtest	Marchenko-Pastur filtering · Clean Σ matrix · Performance comparison	LSTM training · XGBoost features · Regime clustering (K-Means)	Full backtest · Risk attribution · Report submission (Aug 12)

Expected Contribution

- Systematic alpha generation framework for correlation risk premium
- Novel RMT-ML pipeline for volatility regime prediction
- Practical tool for Vol desks at multi-strategy hedge funds